Indian Agriculture Crop Trend Analysis

1. Project Overview and Objective

This project focuses on analysing Indian agricultural crop data from 2010–2020 using Excel and Power BI. The dataset includes information such as crop type, state, season, area cultivated, production, yield, rainfall, temperature, fertilizer usage, and pesticide usage. The project aims to identify agricultural trends, production patterns, and factors affecting crop productivity.

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2. Data Sources

- **Source Description and Timeline:** Crop yields of Indian States and UTs from year 1997-2020 • **Domain**: Agricultural

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3. Problem Statement

To analyse crop production patterns across different states, seasons, and years and evaluate how environmental and agricultural factors such as rainfall, temperature, fertilizer usage, and pesticide usage influence crop yield and production.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Crop	text	Crop cultivated
Year	integer	Year of cultivation
State	Text	State name in India
Season	Text	Kharif,rabi,etc..
Area	Decimal	Cultivated area in Hectare
Production	Decimal	Crop Production in Tonnes
Rainfall	Decimal	Annual Rainfall in mm
Fertilizer	Decimal	Fertilizer uses in kg
Pesticide	Decimal	Pesticide uses in kg
Yield	Decimal	Production by area
Temperature	Decimal	Average temperature in degree Celsius

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5. Tools & Technologies

Microsoft Excel:

- Data Cleaning

- Data Transformation
- Missing Value Treatment
- Duplicate Removal
- Pivot Tables

Power BI :

- Data Modelling
- DAX Measures
- Dashboard Creation

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed

Data Cleaning:

- Removed duplicate records
- Treated missing values
- Corrected inconsistent text formats
- Standardized crop and state names Data Transformation:
- Verified Yield calculations
- Created calculated Productions

Filtering and Sorting :

- Organized records by state, crop, and year

Data Modelling Created:

- Fact Table → Agriculture Data
- Dimension Tables:
    - Dim Crop
    - Dim State
    - Dim Season
    - Dim Year

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables



- **Calculated Columns & DAX Measures:**
- Total cultivation Area = SUM ('Fact agriculture table'[Area(hectare)])

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Total crops = DISTINCTCOUNT ('Fact agriculture table'[Crop])
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Total production = SUM ('Fact agriculture table'[Production (Tones)])
Total state = COUNTA ('Fact agriculture table'[Season])
Yield Status = IF ('Fact agriculture table'[Yield]>=2,"High Yield”, if ('Fact agriculture table'[Yield]<1,"Low Yield”, “Average Yield"))
Average Rainfall = AVERAGE ('Fact agriculture table'[Annual Rainfall(mm)])
Average yield = AVERAGE ('Fact agriculture table'[Yield])
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8. Analysis and Visualizations (Power BI)

Dashboard 1: Production & Yield Analysis

KPI Cards

- Total Production
- Total Area
- Average Yield
- Total crops

Charts

- Production Trend by Year (Line Chart)
- Production by crops (column Chart)
- Production by Season (Donut Chart)
- Yield by Crop (funnel Chart)

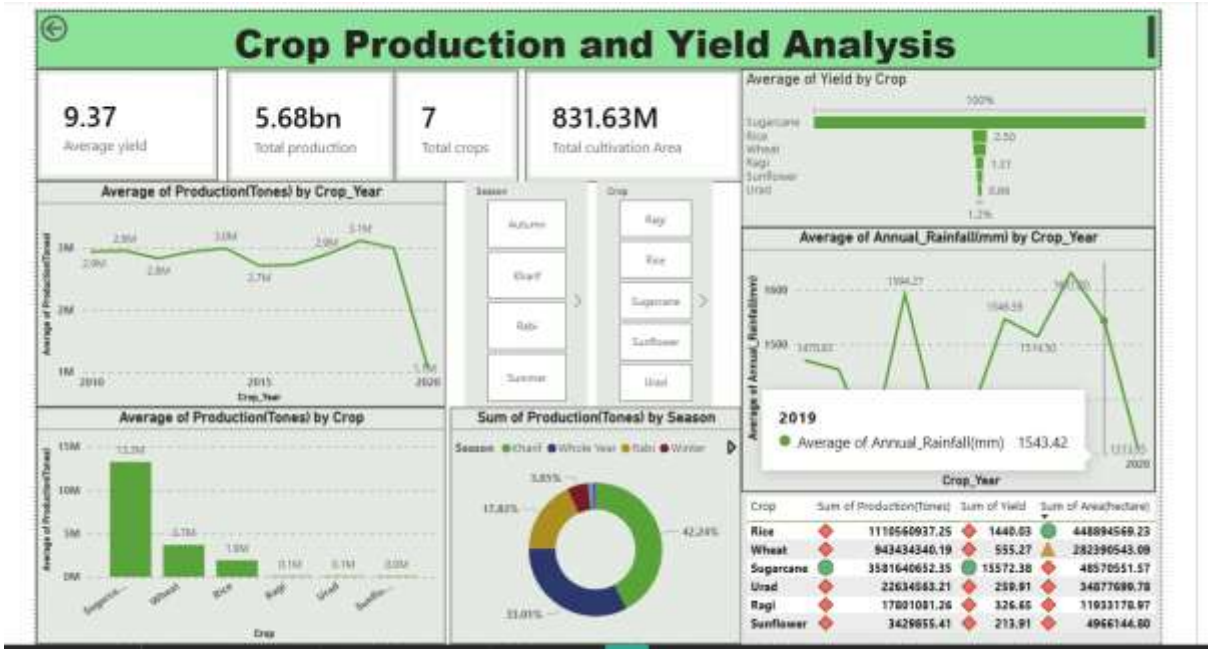
Dashboard 2: Weather & Resource Analysis

Charts

- Rainfall vs Season (Donut Chart)
- Temperature vs Production by Year (line and clustered column Chart)
- Fertilizer vs crop (Ribbon Chart)
- State-wise Production (Tree map Chart)

Interactive Features

- State Slicer
- Crop Slicer
- Year Slicer
- Season Slicer



Key Insights:

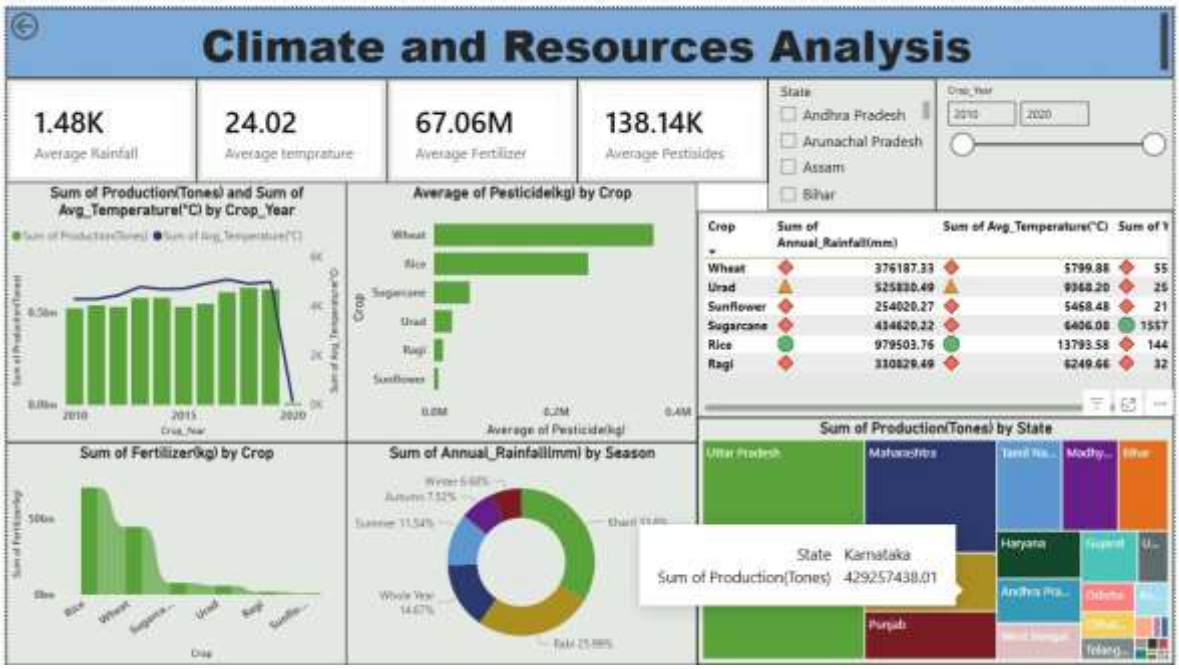
1.sugarcane contributed the largest share of total production.

2.Uttarpradesh is the leading producer among all states.

3.Production has shown an increasing trend over the years, but the Year 2020 the sudden dip was happend.so we have to considered it.

4.Kharif season contributes the highest share of total production.

5.Crop yield generally increases when rainfall is within an optimal range. Excessive or insufficient rainfall is associated with lower production levels.



9. Insights & Conclusions

Descriptive Analysis

- Sugarcane and Wheat contributed the highest share of production.
- Kharif season recorded the highest production.
- Production increased steadily from 20[1] to 2019, but there is a dip after 2019 due to poor Rainfall and High Heat.

Diagnostic Analysis

- States with higher rainfall generally achieved better yields.
- Balanced fertilizer usage positively impacted production.
- Extreme temperatures negatively affected crop productivity.

Predictive Analysis

- Production is expected to continue growing in high-performing states.
- Crops with increasing yield trends may contribute more to future output.

Prescriptive Analysis

- Optimize fertilizer usage based on crop requirements.
- Promote climate-resilient farming practices.
- Focus on improving productivity in low-performing states.

[10.]Conclusions

The integration of Excel and Power BI enabled effective end-to-end agricultural data analysis. The project successfully identified crop production trends, yield patterns, and environmental factors affecting agricultural productivity. The interactive dashboard provides valuable insights that can support farmers, and agricultural organizations in making informed decisions and improving overall agricultural performance.